Extended Answer Questions

Marks will be awarded for working. There are 3 questions in all, all with multiple parts. Show the working in the space provided below each question.

Question 16 **Points: 13**

This question is about differential equations.

(a) For the unknown $u(x,t)$ over $\mathbb{R}\times\mathbb{R}$, solve

$$u_{tt} - 4u_{xx} = 0, \quad u(x,0) = \sin x, \quad u_t(x,0) = \cos(2x).$$

(b) For the unknown $u(x,y)$ over $[0,\pi]\times[0,2\pi]$, solve

$$u_{xx} + u_{yy} = 0, \quad u(x,0) = u_x(0,y) = u_x(\pi,y) = 0, \quad u(x,2\pi) = -\cos(2x).$$

(c) For the unknown $y(x)$, solve

$$y'' - 2y' + y = e^x, \quad y(0) = 0, \quad y(1) = 1.$$

Show your working in the space provided below.

Solution : (a) D'Alembert's Solution.

$f(x) = \sin x$, $g(x) = \cos(2x)$.

Take $c = 2$.

$$u(x,t) = \frac{1}{2}\left(f(x+ct) + f(x-ct)\right) + \frac{1}{2c}\int_{x-ct}^{x+ct} g(s)\,ds$$

$$= \frac{1}{2}\left(\sin(x+2t) + \sin(x-2t)\right) + \frac{1}{4}\int_{x-2t}^{x+2t} \cos(2s)\,ds$$

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$$= \frac{1}{2}\left(\sin(x+2t) + \sin(x-2t)\right) + \frac{1}{8}\left(\sin(2x+4t) - \sin(2x-4t)\right)$$

(b) Step 1: let $u(x,y) = X(x)\cdot Y(y)$.

Plug into the PDE: $X''\cdot Y = -X\cdot Y''$

$$\frac{X''}{X} = -\frac{Y''}{Y} = -\lambda \leftarrow \text{constant}.$$

Step 2: $0 = u(x,0) = X(x)Y(0)$. $Y(0)=0$.

$0 = u_x(0,y) = X'(0)Y(y)$. $X'(0)=0$.

$0 = u_x(\pi,y) = X'(\pi)Y(y)$. $X'(\pi)=0$.

$u(x,2\pi) = -\cos(2x)$ for later.

For $X(x)$, $\begin{cases} X''+\lambda X=0 \\ X'(0)=X'(\pi)=0 \end{cases}$

Eigenvalue Problem (b) with $L=\pi$.

$\lambda_k = \frac{k^2\pi^2}{\pi^2} = k^2$ $\quad k=0,1,2,\dots$

eigenfunction $\cos\left(\frac{k\pi}{\pi}x\right) = \cos(kx)$

Step 3: for $Y(y)$

$\begin{cases} Y_k'' - \lambda_k Y_k = 0 \\ Y_k(0)=0 \end{cases}$ $\quad \lambda_k = k^2$, $k=0,1,\dots$

For $k=0$, $Y_0''=0$. $Y_0(y) = c_1 + c_2 y$. (conti. Page 7)

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(cont. from Page 11)

$Y_0(0) = C_1 = 0.$

$Y_0(y) = Cy.$

For $k = 1, 2, \cdots,$

$$Y_k'' - k^2 Y_k = 0.$$

$$r^2 - k^2 = 0. \quad r = \pm k.$$

$$Y_k = A_k e^{ky} + B_k e^{-ky}$$

$$0 = Y_k(0) = A_k + B_k.$$

So $Y_k = A_k e^{ky} - A_k e^{-ky}.$

The formal general solution is

$$u(x,y) = Cy + \sum_{k=1}^{\infty} A_k (e^{ky} - e^{-ky}) \cos(kx).$$

Step 4:

$$-\cos(2x) = u(x, 2\pi)$$

$$= 2\pi C + \sum_{k=1}^{\infty} A_k (e^{2k\pi} - e^{-2k\pi}) \cos(kx)$$

So $\begin{cases} 2\pi C = 0 \\ A_2(e^{4\pi} - e^{-4\pi}) = -1 \\ A_k(e^{2k\pi} - e^{-2k\pi}) = 0, \ k \neq 2 \end{cases}$

$\begin{cases} C = 0 \\ A_k = 0, \ k \neq 2 \\ A_2 = -\dfrac{1}{e^{4\pi} - e^{-4\pi}}. \end{cases}$

So the solution is

$$u(x,y) = -\frac{e^{2y} - e^{-2y}}{e^{4\pi} - e^{-4\pi}} \cos(x)$$

(c) is on Page 3.

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Q16. (c) $y'' - 2y' + y = e^x$

$y'' - 2y' + y = 0$

$r^2 - 2r + 1 = 0$

$r = 1$ repeated

For a particular solution,

try $y_p(x) = Ax^2e^x$.

$y_p'(x) = 2Axe^x + Ax^2e^x$

$y_p''(x) = 2Ae^x + 4Axe^x + Ax^2e^x$

Plug into the ODE:

x^2e^x: $A - 2A + A = 0$ ✓

xe^x: $4A - 4A = 0$ ✓

e^x: $2A = 1$. $A = \frac{1}{2}$.

$y_p(x) = \frac{1}{2}x^2e^x$.

The general solution is

$y(x) = \frac{1}{2}x^2e^x + c_1e^x + c_2xe^x$

$0 = y(0) = c_1$, $c_1 = 0$

$1 = y(1) = \frac{1}{2}e + c_2e$

$c_2 = \frac{1}{e} - \frac{1}{2}$.

So the solution is

$y(x) = \frac{1}{2}x^2e^x + (\frac{1}{e} - \frac{1}{2})xe^x$

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Question 17 **Points: 9**

This question is about integration.

(a) Calculate the following integrals.

(a.1) For $V = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq 9\}$ in Cartesian coordinates,

$$\iiint_V \frac{\sin(x^2 + y^2 + z^2)}{\sqrt{x^2 + y^2 + z^2}}\, dxdydz$$

(a.2) For $R = \{(x, y) \mid x \geq 0, y \geq 0, x + y \leq 1\}$ in Cartesian coordinates,

$$\iint_R x\, dxdy.$$

(b) Consider $S = \{(x, y, z) \mid x^2 + y^2 + z^2 = 25,\ 0 \leq z \leq 4\}$ and the vector field $\vec{F}(x, y, z) = (y, -z, x)$. Applying Stokes' Theorem or not, calculate

$$\iint_S (\nabla \times \vec{F}) \cdot \vec{n}\, dS$$

where $\vec{n}$ is the unit normal vector field of S pointing in the direction of leaving the origin.

Show your working in the space provided below.

Solution: (a)

(a.1) under spherical coordinate.

$V = \{(\rho, \varphi, \theta) \mid \rho \leq 3\}$.

$$\iiint_V \frac{\sin(x^2+y^2+z^2)}{\sqrt{x^2+y^2+z^2}}\, dx\,dy\,dz$$

$$= \iiint_V \frac{\sin(\rho^2)}{\rho} \cdot \rho^2 \sin\varphi\, d\rho\, d\varphi\, d\theta$$

$$= \int_0^3 \sin(\rho^2) \cdot \rho\, d\rho \cdot \int_0^\pi \sin\varphi\, d\varphi \cdot \int_0^{2\pi} d\theta$$

$(\mu = \rho^2)$

$$= \frac{1}{2} \int_0^9 \sin\mu\, d\mu \cdot (-\cos\varphi)\Big|_0^\pi \cdot 2\pi$$

$$= \frac{1}{2}(1 - \cos 9) \times 2 \times 2\pi$$

$$= 2\pi(1 - \cos 9)$$

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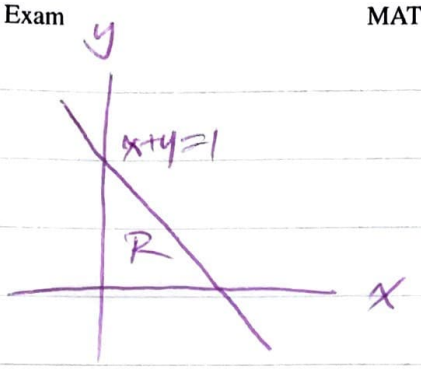
(conti. on Page 5)

(a.2)



$$\iint_R x\,dx\,dy = \int_0^1 \int_0^{1-x} x \; dy \; dx$$

$$= \int_0^1 x \int_0^{1-x} dy \; dx$$

$$= \int_0^1 x(1-x)\,dx$$

$$= \frac{1}{2}x^2 - \frac{1}{3}x^3 \Big|_0^1$$

$$= 1/6$$

(Or $\iint_R x\,dx\,dy = \int_0^1 \int_0^{1-y} x \; dx \; dy$

$$= \int_0^1 \frac{1}{2}x^2 \Big|_0^{1-y} dy$$

$$= \int_0^1 \frac{1}{2}(1-y)^2\,dy$$

$$\overset{1-y=u}{=} -\int_1^0 \frac{1}{2}u^2\,du = 1/6.)$$

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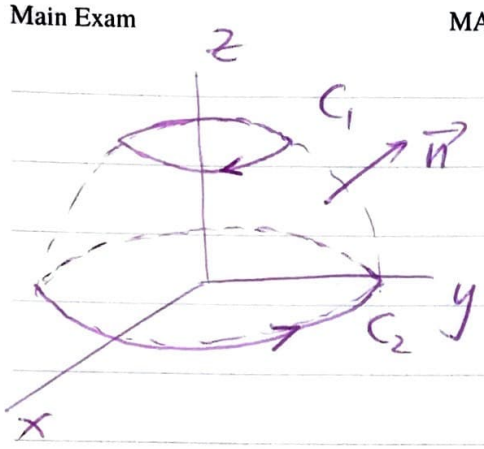
(Conti. on Page 7)

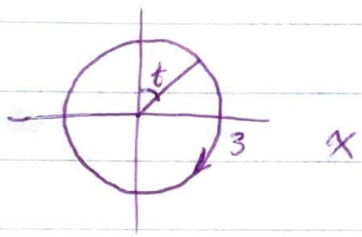
(b)



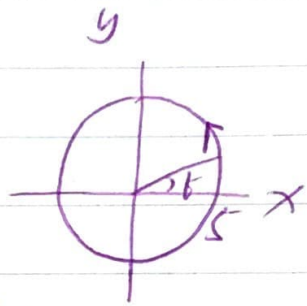
So $\iint_S (\nabla \times \vec{F}) \cdot \vec{n}\, ds$

$= \int_{C_1+C_2} = -16\pi.$

C_1 :



C_2 :



(view from positive z-axis)

C_1: $\vec{r}_1(t) = (3\sin t, 3\cos t, 4)$, $t \in [0, 2\pi]$

C_2: $\vec{r}_2(t) = (5\cos t, 5\sin t, 0)$, $t \in [0, 2\pi]$.

$$\int_{C_1} \vec{F} \cdot d\vec{r}_1 = \int_0^{2\pi} (3\cos t, -4, 3\sin t) \cdot (3\cos t, -3\sin t, 0)\, dt$$

$$= \int_0^{2\pi} \Big(9\underbrace{\cos^2 t}_{\frac{1}{2}(1+\cos 2t)} + 12\underbrace{\sin t}_{\sin t\big|_0^{2\pi}}\Big)\, dt = \tfrac{9}{2} \times 2\pi = 9\pi$$

$$\int_{C_2} \vec{F} \cdot d\vec{r}_2 = \int_0^{2\pi} (5\sin t, 0, 5\cos t) \cdot (-5\sin t, 5\cos t, 0)\, dt$$

$$= \int_0^{2\pi} \Big(-25\underbrace{\sin^2 t}_{\frac{1}{2}(1-\cos(2t))}\Big)\, dt = -\tfrac{25}{2} \times 2\pi = -25\pi$$

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Question 18 **Points: 8**

This question is comprehensive.

(a) Let $H(x, y) = 2x^2 - 2xy + 2y^2$. Consider the vector field

$$\vec{F}(x, y) = (H_y, H_x).$$

The curve $\vec{r}(t) = (x(t), y(t))$ satisfies

$$\vec{r}\,'(t) = \vec{F}(\vec{r}(t)).$$

(a.1) Write down the 1st order ODE system for the unknown $\begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$ from $\vec{r}(t)$.

(a.2) Solve the first order ODE system from (a.1), then get the curve $\vec{r}(t)$ starting from $(2, 1)$, in other words, $\vec{r}(0) = (2, 1)$.

(b) $F(t)$ is a smooth function of variable t. Consider the following integral equation of unknown $y(x)$,

$$y = \int_0^y F(t)dt + \sin x.$$

(b.1) Rewrite the integral equation as a first order ODE of unknown $y(x)$ by taking derivative.

(b.2) If $F(0) = -1$, for the curve of the particular solution of the ODE from $(b.1)$ satisfying $y(0) = 0$, find its tangent line at the origin in the xy-plane $\mathbb{R}^2$.

(c) In $\mathbb{R}^2$, consider a conservative vector field $\vec{F}(x, y) = (f(x, y), g(x, y))$. We define its potential $\Phi(x, y)$ as follows. At the point (a, b), we set $\Phi(a, b) = 0$. Then for any other point (x, y), we pick a curve C from (a, b) to (x, y), and set $\Phi(x, y) = \int_C \vec{F} \cdot d\vec{r}$ which is independent of the choice of such curve C because $\vec{F}$ is conservative. Show $\Phi_x(a, b) = f(a, b)$ and $\Phi_y(a, b) = g(a, b)$ through the following steps.

(c.1) For the point $(a + s, b)$, consider the curve C_s, $\vec{r}(t) = (a + ts, b)$ for $t \in [0, 1]$. Calculate $\Phi(a + s, b) = \int_{C_s} \vec{F} \cdot d\vec{r}$.

(c.2) Using the result from (c.1), calculate $\Phi_x(a, b)$ as $\frac{d\Phi(a+s,b)}{ds}$ at $s = 0$ to see it's equal to $f(a, b)$.

(c.3) Similarly, show $\Phi_y(a, b) = g(a, b)$.

Show your working in the space provided below.

Solution : (a) $\vec{F}(x,y) = (H_y, H_x)$

$$= (-2x+4y,\ 4x-2y)$$

$$\vec{r}\,' = (x', y') = (-2x+4y,\ 4x-2y).$$

(a.1)

$$\begin{bmatrix} x(t) \\ y(t) \end{bmatrix}' = \begin{pmatrix} -2x+4y \\ 4x-2y \end{pmatrix} = \begin{pmatrix} -2 & 4 \\ 4 & -2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$$

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(Conti. on Page 15)

$A = \begin{pmatrix} -2 & 4 \\ 4 & -2 \end{pmatrix}$ Calculate e^{At}.

$$|\lambda I - A| = \begin{vmatrix} \lambda+2 & -4 \\ -4 & \lambda+2 \end{vmatrix} = (\lambda+2)^2 - 16 = 0.$$

$\lambda + 2 = \pm 4$. $\lambda = 2$ or -6.

$\lambda = 2$: $\begin{pmatrix} 4 & -4 \\ -4 & 4 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. Pick $\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

$\lambda = -6$: $\begin{pmatrix} -4 & -4 \\ -4 & -4 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. Pick $\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

$$A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 0 & -6 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}^{-1}$$

$$e^{At} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} e^{2t} & 0 \\ 0 & e^{-6t} \end{pmatrix}\underbrace{\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}^{-1}}$$

$\vec{r}(t) = e^{At}\vec{r}(0)$ $\quad (e^{A\cdot 0} = I_2)$

$$\text{So } \vec{r}(t) = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} e^{2t} & 0 \\ 0 & e^{-6t} \end{pmatrix}\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}\begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} e^{2t} & e^{-6t} \\ e^{2t} & -e^{-6t} \end{pmatrix}\begin{pmatrix} \frac{3}{2} \\ \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{3}{2}e^{2t} + \frac{1}{2}e^{-6t} \\ \frac{3}{2}e^{2t} - \frac{1}{2}e^{-6t} \end{pmatrix}$$

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(conti. on Page 13)

(b) $y(x) = \int_0^{y(x)} F(t)\,dt + \sin x$

Taking x - derivative:

(b.1) $y'(x) = F(y) \cdot y'(x) + \cos x.$

Let $x = 0$:

$y'(0) = F(y(0)) \cdot y'(0) + \cos 0$

$y(0) = 0.$

$y'(0) = F(0) \cdot y'(0) + 1$

$F(0) = -1.$

$y'(0) = \frac{1}{2}.$

$$\boxed{= \int_0^1 \left(f(a+ts, b) + s \cdot t\, f_x(a+ts, b) \right) dt \Big|_{s=0} = \int_0^1 f(a,b)\,dt = f(a,b)}$$

The tangent line is $\boxed{y(x) = \frac{1}{2} x.}$ at $(0,0)$

(c).

(c.3) Consider $\Phi(a, b+s)$ and $\vec{r}(t) = (a, b+ts)$ $t \in [0,1]$

$\vec{r}(t) = (a+ts, b)$

(c.1) $\Phi(a+s, b) = \int_{C_s} \vec{F} \cdot d\vec{r} = \int_0^1 (f(a+ts, b), g(a+ts, b)) \cdot (s, 0)\,dt$

$= \int_0^1 s f(a+ts, b)\,dt$

(c.2) $\left. \frac{d\Phi(a+s, b)}{ds} \right|_{s=0} = \frac{d}{ds} \int_0^1 s f(a+ts, b)\,dt$